

Code:

```
#include <stdio.h> #include <stdlib.h>

struct Node { int data; struct Node* link; };

struct Queue { struct Node *front, *rear; }; typedef struct Queue Queue;

void enQueue(Queue q, int value) { struct Node temp = (struct
Node*)malloc(sizeof(struct Node)); temp->data = value; temp->link = NULL; if (q->front
== NULL) { q->front = temp; } else { q->rear->link = temp; } q->rear = temp; q->rear->link =
q->front; }

int deQueue(Queue* q) { if (q->front == NULL) { return -1; } int value; if (q->front == q-
>rear) { value = q->front->data; free(q->front); q->front = NULL; q->rear = NULL; } else
{ struct Node* temp = q->front; value = temp->data; q->front = q->front->link; q->rear-
>link = q->front; free(temp); } return value; }

void displayQueue(struct Queue* q) { struct Node* temp = q->front; printf("Elements in
Circular Queue are: "); if (q->front != NULL) { while (temp->link != q->front) { printf("%d ",
temp->data); temp = temp->link; } printf("%d\n", temp->data); } else { printf("\n"); } }

int main() { int n; if (scanf("%d", &n) != 1) return 0;

Queue* q = (Queue*)malloc(sizeof(Queue));
q->front = q->rear = NULL;

for (int i = 0; i < n; i++) {
    int val;
    scanf("%d", &val);
    enQueue(q, val);
}

displayQueue(q);
printf("Deleted value = %d\n", deQueue(q));
printf("Deleted value = %d", deQueue(q));
displayQueue(q);

return 0;

}
```

Input:

10
1 2 3 4 5 6 7 8 9 10

Output:

Elements in Circular Queue are: 1 2 3 4 5 6 7 8 9 10

Deleted value = 1

Deleted value = 2 Elements in Circular Queue are: 3 4 5 6 7 8 9 10